SpillThePaper

Deliverable 2: Model Architecture and Empirical Evaluation

AI Course Project, Track A (Application Development)

Minahil Basalat (24L-3082) Iman Abid (24L-0707) Khadija Rao (24L-3042)

FAST-NUCES

Abstract

SpillThePaper aims to turn historical university exam papers into evidence-based exam preparation. Deliverable 1 built the data foundation. This report covers Phase 3 (model architecture and implementation) and Phase 4 (training and empirical evaluation). Because Deliverable 1 produced no labels, we first define a supervised task: given a subject, a topic and a target exam, predict whether the topic appears in that exam, using only papers from earlier years. We build a leakage-safe dataset of 2,290 rows (28 subjects, 220 topics), split it by time, and compare three learned models (logistic regression, random forest and a PyTorch multi-layer perceptron) with four non-learned baselines. On the test split the learned models reach an AUC-ROC of 0.84 to 0.85. This is *not* clearly better than the rule "the topic appeared in the previous year" (0.83): all intervals overlap. The learned models are clearly more precise than the rule of Deliverable 1 ("appeared at least five times"), which flags 81% of all rows. Quizzes and the subject Computer Networks are the weakest cases, probabilities are overestimated, and topics that never appeared before cannot be evaluated at all because the test split contains none. Every number in this report comes from a run of the code in the repository (`https://github.com/Minahilbasalat/SpillThePaper`).

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1 Introduction and Scope

Students at FAST-NUCES have access to many past quizzes, mid-terms and finals, but the collection is large, unstructured and inconsistent. SpillThePaper's long-term goal is to estimate which syllabus areas are likely to appear in an upcoming exam, to generate a tentative paper, and to personalise practice. It is a study aid and does not guarantee what will appear in an actual exam.

Deliverable 1 defined the problem and collected and cleaned 3,567 files from a public student-maintained archive, leaving 2,874 clean exam papers, of which 1,823 have readable text. It produced no question-level information and no prediction target. This report adds both and then trains and evaluates models. It follows the structure of the Deliverable 2 guide (Table 1). The data scripts of Deliverable 1 are not re-run: the committed tables are the inputs of this work.

Deliverable 2 requirement	Where it is covered
Framework selection	Section 3.1 (Scikit-Learn and PyTorch)
Design: layers, loss function, optimizer	Sections 3.4 and 3.5, Figure 3
Hyperparameter table and justification	Section 3.6, Table 6
Validation strategy	Section 4.1 (time-based train/validation/test split)
Precision, recall, F1, AUC-ROC and why	Sections 4.2 and 5
Error analysis	Section 6
Bias check	Section 7
Architecture diagrams, loss and accuracy curves	Figures 1, 3, 5
Repository, `requirements.txt`, weights link	Section 9
Contribution table (authors and modules)	Section 10

Table 1: Coverage of the Deliverable 2 requirements.

Figure 1 shows the pipeline of this deliverable.

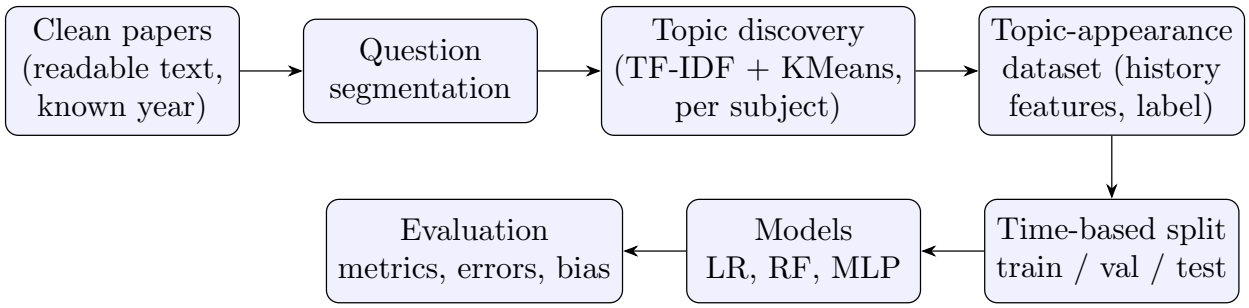


Figure 1: Pipeline of Phases 3 and 4. LR = logistic regression, RF = random forest, MLP = multi-layer perceptron.

2 Problem Formulation and Dataset

2.1 Supervised task

One row of the dataset is a (subject, topic, target exam) triple, where a target exam is a (year, assessment type) pair of a subject. The label is 1 if at least one paper of that subject, year and type contains a question assigned to the topic, and 0 otherwise. The model estimates the probability that a topic appears, which fits the uncertainty argument of the Deliverable 1 AI justification. All features use only papers from strictly earlier years than the target exam.

2.2 Question segmentation

The 1,638 clean papers with readable text and a known year (789 quizzes, 532 mids, 317 finals; 105 subjects) are split into questions by a rule-based segmenter that tries three strategies in order: explicit markers (`Question 1`, `Q2`), numbered items, and a paragraph fallback. Worked solutions, grading tables and paper-header paragraphs are removed. The result is 7,780 questions from 1,575 papers (mean 4.9, median 3 per paper): 1,001 papers were split on markers, 252 on numbering and 322 by the paragraph fallback. The other 63 papers yielded no question after filtering. Segmentation was checked on random samples and by unit tests, but its accuracy was not measured against a manually labelled set.

2.3 Topic discovery

Topics are discovered per subject, because a topic is only meaningful inside one course. For each subject we fit a TF-IDF representation (unigrams and bigrams, 2,000 features, English and exam-specific stop words) and KMeans clustering with $K \in [4, 10]$ chosen by cosine silhouette score. Topics are fitted on questions from years up to 2022 only; later questions are assigned to the nearest existing topic, so topic definitions never use future papers.

A subject needs at least 6 papers from at least 3 distinct years and 30 questions in the training period. Of 105 subjects, 28 qualify; 77 are reported as low-confidence and excluded (54 had fewer than 6 training papers, 21 fewer than 3 training years, one a vocabulary that was too small and one fewer than 30 questions). The 28 subjects have 220 topics (median $K = 9$) and 4,861 of the 7,780 questions (62.5%) are assigned to a topic. Silhouette scores are low (median 0.097, range 0.042 to 0.188), as expected for short exam questions with overlapping vocabulary. Most topics are interpretable (for example “circuit, switching” in Computer Networks, “fitness, chromosome, population” in Artificial Intelligence), but some clusters capture generic wording instead of course content.

2.4 Features

Table 2 lists the features. `years_since_last` is missing when a topic has never appeared before and is imputed on the training split only (Section 3.2).

Feature	Meaning
`prior_papers`, `prior_years`	papers and distinct years of the subject before the target year
`topic_count_prior`, `topic_rate_prior`	appearances of the topic before, and that count divided by `prior_papers`
`years_since_last`	years since the topic last appeared
`appeared_last_year`	1 if the topic appeared in the year before the target year
`recent_rate`	appearance rate over the previous two years
`type_rate_prior`	appearance rate among earlier papers of the same assessment type
`target_type`	quiz, mid or final (one-hot encoded)

Table 2: Model features. A target exam needs at least 3 earlier papers from at least 2 earlier years.

2.5 Data split and leakage checks

The split is by time. Training uses target years 2014 to 2022, validation 2023 and testing 2024 to 2026. The years were chosen from paper counts before any model was trained. Table 3 and Figure 2 summarise the result.

Split	Rows	Positives	Positive rate	Target exams	Subjects
Train (2014–2022)	1,421	672	0.473	166	28
Validation (2023)	301	118	0.392	36	22
Test (2024–2026)	568	197	0.347	67	21

Table 3: Size and class balance of each split (2,290 rows in total).

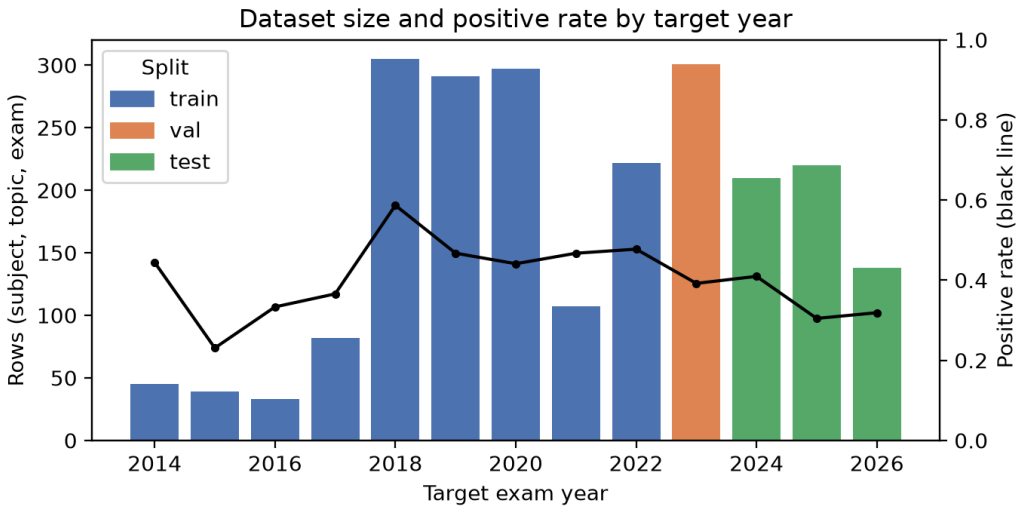


Figure 2: Rows per target year, coloured by split, with the positive rate (black line).

Two checks protect against leakage. A unit test changes only the target-year papers and verifies that every feature stays identical while the label changes. As a sanity check we also computed

the AUC-ROC of single raw features: `topic_rate_prior`, `recent_rate` and `topic_count_prior` reach 0.66 to 0.82 on validation and 0.79 to 0.82 on test, far below the near-perfect values that a leak would produce. These values also show that a simple frequency feature is already a strong baseline.

Data-side setting	Tried	Final	Reason
Min. training papers per subject	10, 6	6	10 gave 26 eligible subjects; 6 gave 28
Min. training questions	40, 30	30	enough for stable clustering
Min. training years	3	3	recency features need history
Topics per subject K	4 to 10 by silhouette	4–10	no single K suits every subject
Stop words	English; English + exam + header words	the latter	header words formed meaningless clusters
Min. prior papers / years per target exam	3 / 2	3 / 2	history features would be empty
Recent window	2 years	2	syllabi change over time

Table 4: Data-side settings (also in `docs/data_settings.md`).

3 Phase 3: Model Architecture and Training

3.1 Framework choice

We use two industry-standard frameworks. Scikit-Learn 1.9.1 provides logistic regression and a random forest, because it offers reliable, well-tested implementations and a small, reproducible tuning interface. PyTorch 2.15.0 (CPU) provides the multi-layer perceptron (MLP), because it gives explicit control over the layers, the loss function, the optimizer and the training loop (mini-batches, early stopping, restoring the best weights). The code runs on Python 3.14 with a fixed seed of 42 unless stated otherwise.

3.2 Preprocessing

All preprocessing statistics are fitted on the training split only (1,421 rows) and applied unchanged to validation and test. A unit test checks that changing validation rows does not change any fitted statistic and that no missing value remains after the transform.

- **Missing values.** `years_since_last` is missing for topics never seen before (128 of 1,421 training rows). It is filled with the training maximum plus one (10.0), and a binary column `never_seen` marks the filled rows.
- **Encoding.** `target_type` is one-hot encoded with the categories seen in training (final, mid, quiz).
- **Scaling.** The eight numeric features are standardised with the training mean and standard deviation.
- **Excluded columns.** Subject, topic id and topic label are not features, so the models cannot memorise subjects or topics.

The result is 12 input features: 8 numeric, 1 indicator and 3 one-hot columns.

3.3 The models

Table 5 lists the learned models. The raw feature `topic_rate_prior` (the share of earlier papers in which the topic appeared), used directly as a score without any learning, is an additional reference.

Model	Framework	Selected settings
Logistic regression	Scikit-Learn	$C = 0.01$
Random forest	Scikit-Learn	300 trees, maximum depth 5, minimum 5 samples per leaf
MLP	PyTorch	12–16–8–1, dropout 0.4, Adam with learning rate 0.01 (Section 3.6)

Table 5: Learned models. All settings were selected on validation AUC-ROC.

For logistic regression we tried $C \in \{0.01, 0.1, 1, 10\}$; the validation AUC-ROC fell slightly from 0.847 to 0.843 as C grew, so stronger regularisation was marginally better. For the random forest we tried 16 settings (trees in $\{100, 300\}$, maximum depth in $\{3, 5, 8, \text{none}\}$, minimum samples per leaf in $\{1, 5\}$). Validation AUC-ROC ranged from 0.789 to 0.827. Forests of unlimited depth with one sample per leaf reached a training AUC-ROC of 0.99 but the lowest validation AUC-ROC (0.79), which is overfitting, so a shallow forest was selected.

3.4 MLP architecture

Figure 3 shows the network. It has two hidden layers and a single output unit:

$$\text{Linear}(12, 16) \to \text{ReLU} \to \text{Dropout}(0.4) \to \text{Linear}(16, 8) \to \text{ReLU} \to \text{Dropout}(0.4) \to \text{Linear}(8, 1).$$

The selected network has 353 trainable parameters ($12 \cdot 16 + 16 + 16 \cdot 8 + 8 + 8 + 1$). The other tested sizes have 961 parameters (32–16) and 2,945 parameters (64–32). A small network suits a dataset of 1,421 training rows with 12 features.

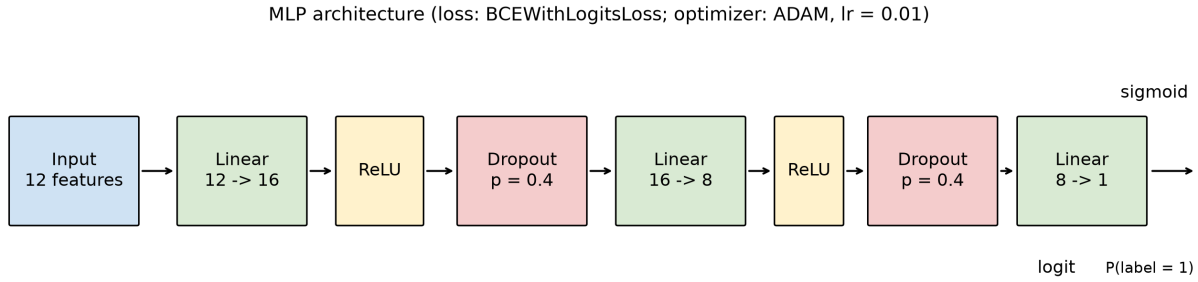


Figure 3: Architecture of the final MLP. The network outputs a logit; a sigmoid turns it into the probability that a topic appears.

Layers. ReLU is a standard, cheap non-linearity. Dropout randomly switches off hidden units during training as a regulariser. The output is a raw logit; the sigmoid is applied only when probabilities are needed.

3.5 Loss function and optimizer

Loss. We use `BCEWithLogitsLoss`, which combines the sigmoid and binary cross-entropy in one numerically stable step and is the natural loss for a binary label with a probability output. The optional `pos_weight` is left off: 47.3% of training rows are positive, so the weight would be only about 1.11 and the classes are nearly balanced.

Optimizer. We use Adam with learning rate 0.01, no weight decay and batch size 64. Adam adapts the step size per parameter and needs little manual tuning. SGD with momentum 0.9 was the alternative in the search. Training stops early when the validation loss has not improved for 15 epochs (at most 200 epochs), and the weights of the best epoch are restored.

3.6 Hyperparameter tuning

We ran a random search over the space in Table 6, which contains 486 combinations. Forty distinct configurations were drawn with a fixed seed. Each configuration was trained with three seeds (42, 43, 44) and ranked by the *mean* validation AUC-ROC over the three seeds, taken at each run's best validation-loss epoch. The test split was never loaded during tuning. All 40 results are stored in `results/tuning_results.csv`.

Parameter	Tested values	Final	Mean val. AUC-ROC per value (configs)	Justification
Learning rate	0.001, 0.003, 0.01	0.01	0.8358 (19), 0.8385 (9), 0.8378 (12)	Value of the best configuration; group means differ by less than 0.003. Its best epoch comes after about 10 epochs on average.
Batch size	32, 64, 128	64	0.8374 (15), 0.8366 (11), 0.8370 (14)	No meaningful difference (range 0.001); 64 is the best configuration's value and gives 23 updates per epoch.
Dropout	0.0, 0.2, 0.4	0.4	0.8359 (21), 0.8393 (9), 0.8374 (10)	Group means favour 0.2, so 0.4 is not clearly better. Dropout 0.0 had the lowest mean, so some regularisation helps.
Hidden sizes	(16,8), (32,16), (64,32)	(16,8)	0.8363 (10), 0.8356 (12), 0.8384 (18)	Group means slightly favour (64,32); (16,8) is the best configuration and the smallest network (353 parameters).
Weight decay	0, 1e-4, 1e-3	0	0.8372 (12), 0.8365 (12), 0.8373 (16)	No visible effect, so the simplest setting is kept.
Optimizer	Adam, SGD (momentum 0.9)	Adam	0.8386 (20), 0.8354 (20)	Adam has a slightly higher mean (gap about 0.003) and needs no momentum setting.

Table 6: Tested hyperparameters, selected values and the mean validation AUC-ROC over all sampled configurations that used each value. Fixed: `pos_weight` off, patience 15, at most 200 epochs.

The selected configuration (number 35) reached a mean validation AUC-ROC of 0.8418 with a standard deviation of 0.0026 over the three seeds. The five best configurations lie between 0.8413 and 0.8418, a gap smaller than the seed-to-seed standard deviation (about 0.002 to 0.003), so the best configuration is only marginally better than its neighbours. Across all 40 configurations the mean validation AUC-ROC ranged from 0.822 to 0.842, so architecture and optimizer matter little on this data. The final configuration is the best single configuration, not a combination of the best value of each parameter. Because random sampling gave unequal numbers of configurations per value (for example, learning rate 0.001 appears in 19 of 40), the per-value means are descriptive only. Figure 4 shows the 40 results.

3.7 Training curves

Figure 5 shows the loss and accuracy of the final MLP (selected configuration, seed 42). Training stopped at epoch 24 and the best validation loss (0.5136) occurred at epoch 9, so the weights of epoch 9 are used. The final run reached a validation AUC-ROC of 0.838, slightly below the three-seed mean of 0.8418, which is ordinary seed-to-seed variation.

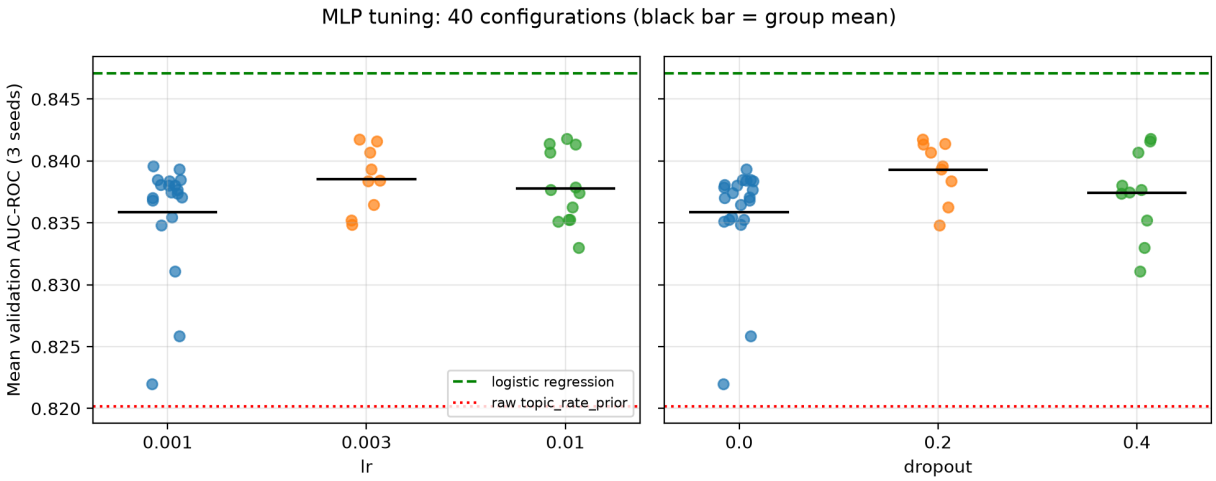


Figure 4: Mean validation AUC-ROC (three seeds) of the 40 tested configurations, grouped by learning rate (left) and dropout (right). Black bars are group means. The dashed line is logistic regression and the dotted line is the raw `topic_rate_prior` feature.

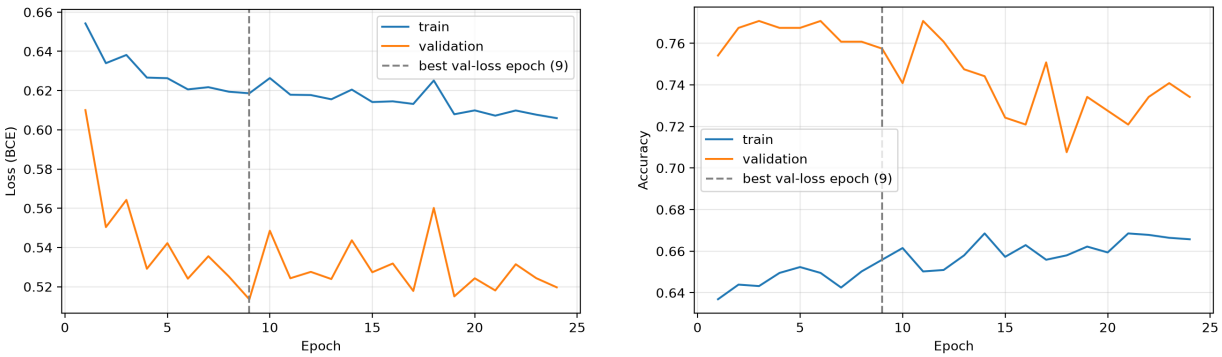


Figure 5: Training and validation loss (left) and accuracy (right) of the final MLP. The dashed line marks the epoch with the lowest validation loss (epoch 9).

The curves do not show classic overfitting. The training loss decreases slowly, while the validation loss is noisy and does not improve clearly after epoch 9. Validation accuracy peaks around 0.77 in the first epochs and varies between 0.71 and 0.75 after epoch 13, which supports stopping early. Training accuracy stays between 0.64 and 0.67; for comparison, always predicting "does not appear" would score 0.527 on the training split and 0.608 on validation. The validation loss is lower than the training loss and validation accuracy is higher than training accuracy. Both are computed with dropout switched off, so dropout does not explain the gap. A likely reason is that the validation split differs from the training split: it has a lower positive rate (39.2% against 47.3%) and covers only 22 subjects and 36 target exams. We did not test this explanation, and the gap should not be read as evidence of better generalisation. The modest training accuracy suggests that the task is hard for these features rather than that the model is too complex.

4 Phase 4: Validation and Metrics

4.1 Validation strategy

We use a chronological train/validation/test split instead of random K-fold cross-validation. A random split would place future exams in the training data, so the model would be tested on years it has effectively already seen. It would also hide the drift of the positive rate over time (47.3%,

39.2%, 34.7%; Table 3), and it does not match the real use, which is to predict an upcoming exam from past ones. The split holds 1,421 / 301 / 568 rows (62% / 13% / 25%), and 166 / 36 / 67 target exams. The validation split is the only set used to choose models, hyperparameters and decision thresholds. The test split is evaluated once, after all models and thresholds were fixed. During tuning, three seeds per MLP configuration measure the seed-to-seed variation.

Decision threshold. A learned model outputs a probability. The threshold that turns it into a yes/no prediction is chosen once on validation (maximum F1; ties go to the smallest threshold) and applied unchanged to the test split. Rule baselines defined by a fixed rule keep that rule as their threshold.

Confidence intervals. Rows of the same target exam (subject, `target_year`, `target_type`) are strongly correlated: they share the same papers, and the label depends on which papers the exam group contains. Treating rows as independent would make intervals too narrow. We therefore resample whole target exams with replacement (1,000 resamples, seed 42) and report 95% percentile intervals. The test split contains only 67 exams and validation only 36, so small metric differences are within noise. Comparing two intervals of the same exams without pairing is conservative: overlapping intervals do not prove that two models are equal, only that this data cannot separate them.

4.2 Metric choice and justification

The classes are roughly balanced but drifting (47.3% positives in training, 34.7% in test), so accuracy alone says little. A student also pays two different costs. A false negative is a topic that appears in the exam but was not recommended; a false positive is a topic studied for nothing. Accuracy hides this trade-off, so we report:

- **Precision**: of the topics we recommend, the share that appear.
- **Recall**: of the topics that appear, the share we recommend.
- **F1-score**: the harmonic mean of precision and recall; it is also the criterion for the threshold.
- **AUC-ROC**: the probability that a random appearing topic is scored above a random non-appearing topic. It does not depend on a threshold, so it measures ranking quality separately from the threshold choice.
- **PR-AUC** (average precision): the area under the precision-recall curve, more informative than AUC-ROC when the positive rate differs between splits.
- **Brier score**: the mean squared error of the predicted probability (lower is better). It checks calibration, which matters because the positive rate drifts. It is not reported for scores that are not probabilities (counts) or for the constant all-positive score.

4.3 Baselines

Four non-learned baselines show whether learning adds anything over simple rules (Table 7).

Baseline	Threshold	Rule or score
Rule: count ≥ 5	fixed rule	predict 1 if the topic appeared at least 5 times before (the "appeared 5 times, so predict it" rule of the Deliverable 1 AI justification)
Rule: appeared last year	fixed rule	predict 1 if the topic appeared in the previous year
Raw topic rate	chosen on validation	score = `topic_rate_prior`
All positive	fixed rule	predict 1 for every row

Table 7: Non-learned baselines.

5 Results

Table 8 gives the validation results, on which models and thresholds were selected. Table 9 and Table 10 give the test results with 95% cluster-bootstrap intervals.

Model	Threshold	Precision	Recall	F1	AUC-ROC	PR-AUC	Brier
Logistic regression	0.470	0.696	0.814	0.750	0.847	0.784	0.179
Random forest	0.470	0.640	0.814	0.716	0.827	0.742	0.174
MLP (PyTorch)	0.484	0.656	0.839	0.736	0.838	0.778	0.170
Rule: count ≥ 5	5.000	0.491	0.932	0.643	0.790	0.697	–
Rule: appeared last year	1.000	0.542	0.551	0.546	0.775	0.673	–
Raw topic rate	0.345	0.687	0.763	0.723	0.820	0.731	0.176
All positive	1.000	0.392	1.000	0.563	0.500	0.392	–

Table 8: Validation results (301 rows, 36 target exams). Thresholds of the learned models and of the raw topic rate maximise F1 on this split; the other rules keep their fixed threshold. This table is therefore optimistic for them.

Model	Precision	Recall	F1
Logistic regression	0.621 [0.550, 0.688]	0.766 [0.698, 0.838]	0.686 [0.632, 0.732]
Random forest	0.595 [0.525, 0.662]	0.858 [0.800, 0.910]	0.703 [0.654, 0.745]
MLP (PyTorch)	0.597 [0.525, 0.665]	0.858 [0.805, 0.913]	0.704 [0.652, 0.749]
Rule: count ≥ 5	0.418 [0.363, 0.473]	0.975 [0.954, 0.991]	0.585 [0.528, 0.637]
Rule: appeared last year	0.579 [0.511, 0.639]	0.802 [0.725, 0.868]	0.672 [0.614, 0.724]
Raw topic rate	0.568 [0.504, 0.631]	0.655 [0.572, 0.739]	0.608 [0.560, 0.651]
All positive	0.347 [0.296, 0.398]	1.000 [1.000, 1.000]	0.515 [0.457, 0.569]

Table 9: Test results, threshold-dependent metrics, with 95% cluster-bootstrap intervals (568 rows, 67 target exams). Thresholds were fixed on validation.

5.1 Reading the results

Ranking quality. On the test split the three learned models reach an AUC-ROC of 0.839 (logistic regression), 0.843 (random forest) and 0.849 (MLP), with intervals of about ± 0.04 that overlap almost completely. They also overlap with the history rules: "appeared last year" reaches 0.827 and "count $\geq$ 5" 0.822. The raw topic rate is lowest at 0.787, and even its interval overlaps with those of the learned models. PR-AUC tells the same story. We therefore cannot claim that any learned model ranks topics better than a one-line rule. The three learned models cannot be

Model	AUC-ROC	PR-AUC	Brier score
Logistic regression	0.839 [0.796, 0.880]	0.756 [0.690, 0.819]	0.173 [0.162, 0.186]
Random forest	0.843 [0.804, 0.881]	0.741 [0.671, 0.812]	0.172 [0.160, 0.186]
MLP (PyTorch)	0.849 [0.811, 0.886]	0.758 [0.694, 0.823]	0.163 [0.150, 0.178]
Rule: count ≥ 5	0.822 [0.783, 0.859]	0.705 [0.641, 0.769]	–
Rule: appeared last year	0.827 [0.787, 0.865]	0.713 [0.652, 0.776]	–
Raw topic rate	0.787 [0.742, 0.830]	0.688 [0.627, 0.747]	0.174 [0.152, 0.198]
All positive	0.500 [0.500, 0.500]	0.347 [0.296, 0.398]	–

Table 10: Test results, threshold-free and probability metrics, with 95% intervals. A constant prediction of the test positive rate (0.347) would score a Brier value of $0.347 \times 0.653 = 0.227$.

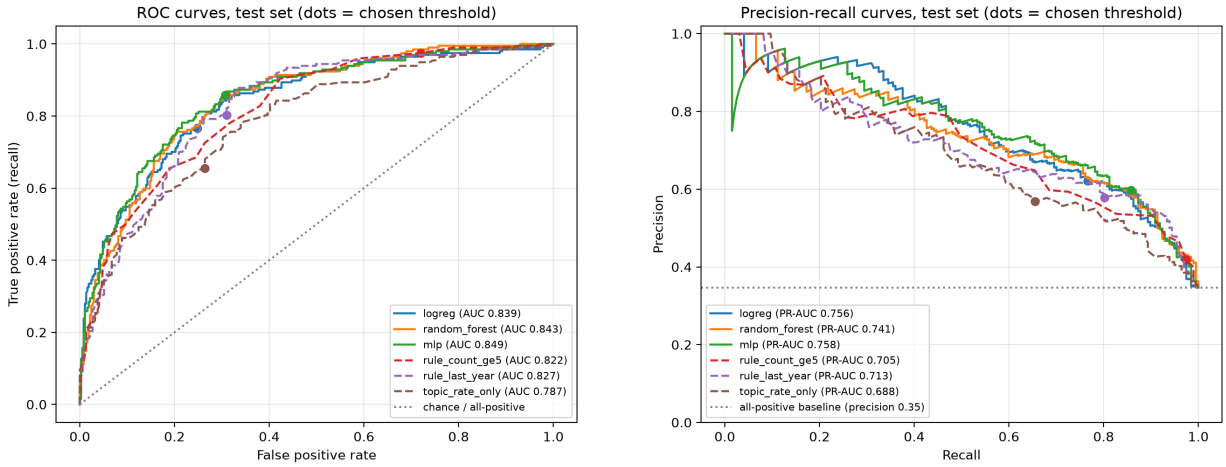


Figure 6: ROC curves (left) and precision-recall curves (right) on the test split.

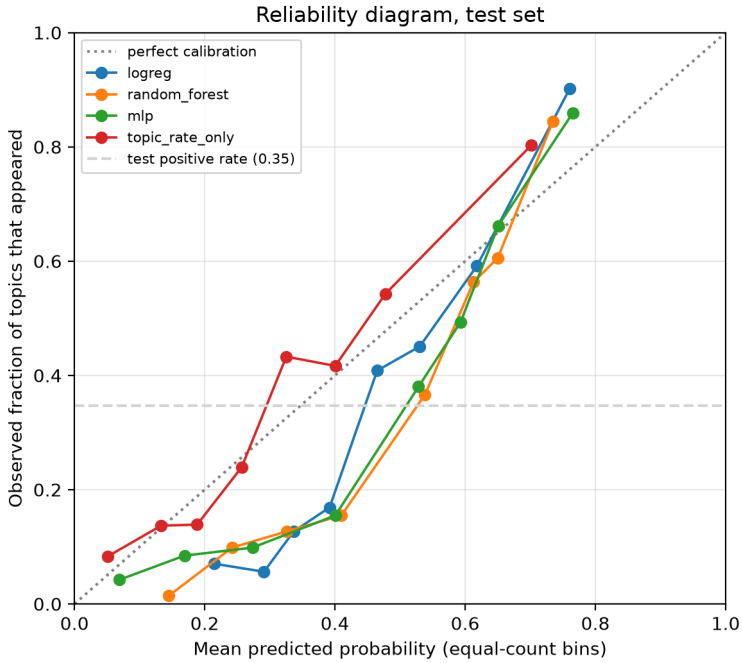


Figure 7: Reliability diagram on the test split: mean predicted probability against the observed fraction of topics that appeared.

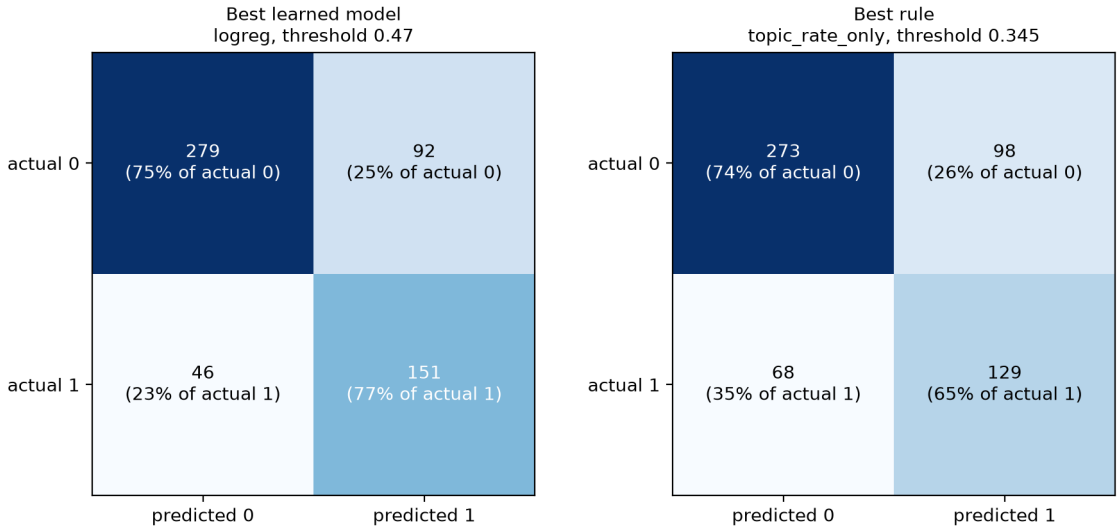


Figure 8: Confusion matrices on the test split: the best learned model (logistic regression) and the best rule (raw topic rate).

told apart from each other either, which agrees with the tuning results (Section 3.6). Only the all-positive baseline (AUC-ROC 0.5) is clearly worse.

Against the rule of Deliverable 1. The rule "appeared at least five times" flags 459 of 568 test rows (81%) as positive. This gives a recall of 0.975 at a precision of only 0.418 (F1 0.585). The learned models flag 43% to 50% of the rows, and their precision (0.595 to 0.621) is clearly higher: the intervals do not overlap. Their recall is lower (0.766 to 0.858). Their F1 (0.686 to 0.704) is clearly higher for the random forest and the MLP, while the interval of logistic regression still overlaps that of the rule. One caveat applies: the learned thresholds were tuned on validation, whereas the rule's threshold of 5 was fixed in advance, so part of this F1 advantage reflects threshold tuning and not better ranking, which the AUC-ROC comparison shows to be similar.

Against the other baselines. "Appeared last year" cannot be separated from any learned model on any metric (F1 0.672 against 0.686 to 0.704). The raw topic rate with a validation-tuned threshold reaches an F1 of 0.608 on test; the intervals of the random forest and the MLP separate from it only narrowly, and the recall of these two models is clearly higher.

Probability quality. All three learned models beat the constant-rate Brier value of 0.227 (MLP 0.163, random forest 0.172, logistic regression 0.173), but they overestimate: the mean predicted probability on test is 0.451 (logistic regression), 0.457 (random forest) and 0.431 (MLP), against an observed rate of 0.347. This matches the drift in the positive rate. The reliability diagram (Figure 7) shows the overestimate mainly for scores below about 0.55; for the highest scores the observed fractions are close to the diagonal or above it. Predicted probabilities should be read as relative rankings, not as calibrated chances.

Validation to test. The AUC-ROC moved from 0.847 to 0.839 (logistic regression), from 0.827 to 0.843 (random forest) and from 0.838 to 0.849 (MLP). Selecting on validation therefore did not produce an optimistic gap larger than the width of the intervals. F1 fell for all three models (for example 0.750 to 0.686 for logistic regression), mainly because precision fell as the positive rate dropped from 39% to 35%.

Summary. The learned models are competitive with, but not demonstrably better than, simple history rules. Their clearest advantage over the rule of Deliverable 1 is precision: they recommend fewer topics for a similar number of hits. This is an honest negative result for the claim that machine learning is necessary on this feature set. It does not settle whether richer, question-level features would change the picture (Section 8).

6 Error Analysis

We analyse the learned model with the highest validation F1, which is logistic regression (0.750; MLP 0.736, random forest 0.716), at its validation threshold of 0.470. The test split is only broken down here and never used for a choice. On the test split this model makes 46 false negatives and 92 false positives (151 true positives, 279 true negatives). For each group Table 11 gives the numbers of rows and exams, the positive rate, the mean predicted score (a miscalibration check), precision, recall, F1, AUC-ROC and the counts of false negatives (FN) and false positives (FP). Figure 9 summarises the groups.

Group	Rows	Exams	Pos. rate	Mean score	Prec.	Rec.	F1	AUC	FN	FP
Assessment type										
final	192	22	0.339	0.467	0.607	0.831	0.701	0.875	11	35
mid	236	29	0.301	0.443	0.604	0.859	0.709	0.892	10	40
quiz	140	16	0.436	0.442	0.679	0.590	0.632	0.734	25	17
Target year										
2024	210	26	0.410	0.466	0.673	0.791	0.727	0.846	18	33
2025	220	26	0.305	0.451	0.560	0.761	0.646	0.821	16	40
2026	138	15	0.319	0.427	0.627	0.727	0.674	0.846	12	19
Earlier papers of the subject										
10-19	92	13	0.293	0.472	0.667	0.963	0.788	0.946	1	13
20+	476	54	0.357	0.447	0.613	0.735	0.668	0.823	45	79
Earlier appearances of the topic										
1-2	47	32	0.043	0.247	0.000	0.000	0.000	0.067	2	0
3-5	114	53	0.053	0.315	0.200	0.167	0.182	0.537	5	4
6+	407	67	0.464	0.513	0.630	0.794	0.703	0.789	39	88

Table 11: Test error analysis of logistic regression by group.

Quizzes are the main weakness. The recall on quizzes is 0.590, against 0.831 for finals and 0.859 for mids, and the AUC-ROC is 0.734 against 0.875 and 0.892. Quizzes make up 25% of the test rows but 25 of the 46 false negatives (54%). Quizzes are short (median 2 estimated questions in Deliverable 1), so one quiz covers only a few of a subject's topics. We did not test whether this explains the gap.

Computer Networks is under-predicted. Table 12 shows that Computer Networks has the highest positive rate (0.588) but a mean score of only 0.405, a recall of 0.574 and a precision of 0.964. It accounts for 20 of the 46 false negatives (43%). Topics such as shortest paths, HTTP/TCP and round-trip time are missed most often (Table 13). The false positives are spread over other subjects (Table 14): PP, Theory of Automata, Database Systems and Design and Analysis of Algorithms.

The same topics appear in both lists in several subjects (for example the PP topics "law, ethics, example, pakistan" and "year, rate, years, value", and the Database Systems topic "customer, number, branch, account"), so in some subjects whether a topic appears in a given exam is only weakly predictable from its history. The most frequent false positive in PP (5 of 6 rows) is a cluster whose keywords ("exam, solve, requirements, framework") look like generic wording rather than syllabus content, which shows how topic noise affects the predictions.

Most of the difficulty lies in frequently seen topics. 407 of the 568 test rows (72%) are topics that appeared at least six times before, and 46.4% of them are positive. The 161 rows of topics seen only one to five times contain only 8 positives (4–5%), yet the model gives them mean

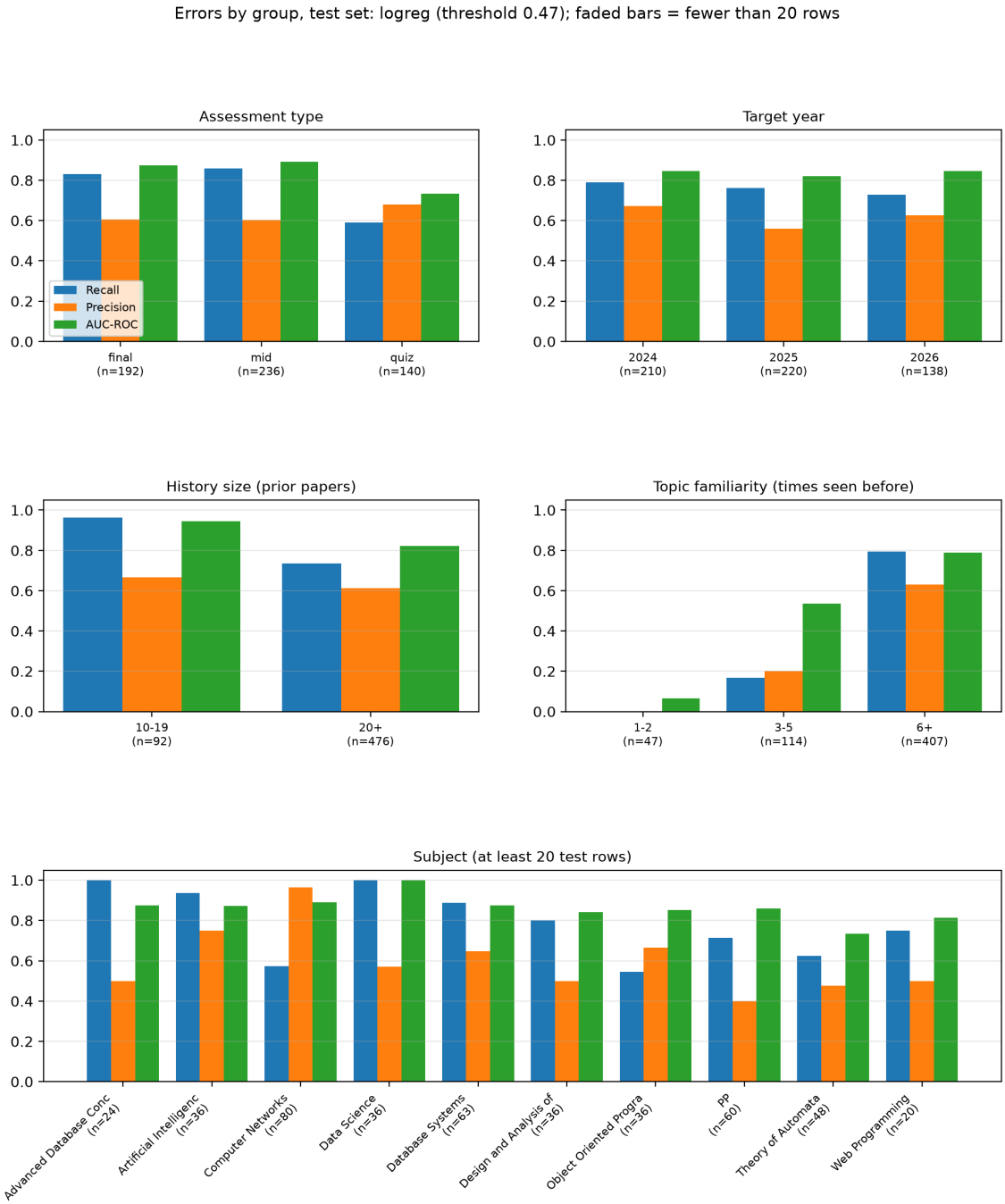


Figure 9: Error analysis by group on the test split.

scores of 0.247 and 0.315 (one value per group), so rare topics are overestimated. The model makes few errors there only because these scores lie below the threshold. The AUC-ROC of these two groups rests on 2 and 6 positive rows and is not meaningful. Among the frequent topics, which hold 88 of the 92 false positives and 39 of the 46 false negatives, the AUC-ROC is 0.789, noticeably lower than the overall 0.839. Part of the overall AUC-ROC therefore comes from the easy contrast between frequent and rare topics.

No cold-start evidence. The test split contains no topic that never appeared before. All 128 such rows are in the training split. Performance on entirely new topics, the case where learning could matter most, is therefore not measured by this evaluation.

Years and history. The AUC-ROC is 0.846, 0.821 and 0.846 for 2024, 2025 and 2026 and the

Subject	Rows	Exams	Pos. rate	Mean score	Prec.	Rec.	F1	AUC	FN	FP
Advanced Database Concepts	24	3	0.333	0.503	0.500	1.000	0.667	0.875	0	8
Artificial Intelligence	36	4	0.444	0.495	0.750	0.938	0.833	0.872	1	5
Computer Networks	80	8	0.588	0.405	0.964	0.574	0.720	0.890	20	1
Data Science	36	4	0.111	0.424	0.571	1.000	0.727	1.000	0	3
Database Systems	63	7	0.429	0.495	0.649	0.889	0.750	0.874	3	13
Design and Analysis of Algorithms	36	4	0.278	0.438	0.500	0.800	0.615	0.842	2	8
Object Oriented Programming	36	4	0.306	0.355	0.667	0.545	0.600	0.851	5	3
PP	60	6	0.233	0.460	0.400	0.714	0.513	0.859	4	15
Theory of Automata	48	6	0.333	0.455	0.476	0.625	0.541	0.734	6	11
Web Programming	20	2	0.200	0.432	0.500	0.750	0.600	0.812	1	3

Table 12: Test error analysis of logistic regression by subject (subjects with at least 20 test rows; 439 of 568 rows).

Subject	Topic keywords	FN / rows
Computer Networks	node, algorithm, shortest, paths	5 / 8
Computer Networks	connection, http, tcp, object	4 / 8
Computer Networks	transmission round, round, rtt, transmission	4 / 8
Computer Networks	frame, sends frame, replies frame, replies	2 / 8
Computer Networks	segment, port, port number, destination port	2 / 8
Computer Networks	circuit, switching, switched, circuit switching	2 / 8
Database Systems	customer, number, branch, account	2 / 7
PP	law, ethics, example, pakistan	2 / 6
PP	year, rate, years, value	2 / 6
Theory of Automata	grammar, false, true, context free	2 / 6

Table 13: Topics with the most false negatives (topic keywords come from the clustering).

Subject	Topic keywords	FP / rows
PP	exam, solve, requirements, framework	5 / 6
Theory of Automata	initial, deterministic, deterministic finite, finite	5 / 6
Database Systems	customer, number, branch, account	4 / 7
Database Systems	update, key, tuple, constraint	4 / 7
Theory of Automata	dfa, required, nfa, table	4 / 6
PP	profession, malicious, reasons, information	3 / 6
PP	law, ethics, example, pakistan	3 / 6
PP	year, rate, years, value	3 / 6
Design and Analysis of Algorithms	recurrence, sum, function, left	3 / 4
Design and Analysis of Algorithms	tree, edge, prove, graph	3 / 4

Table 14: Topics with the most false positives.

intervals overlap (Table 11), so we see no trend over time. Precision is lowest in 2025 (0.560), the year with the lowest positive rate (0.305), which agrees with the drift. Subjects with 20 or more earlier papers have a lower recall (0.735) than subjects with 10 to 19 (0.963), but this is confounded with the subject, since the large subjects include Computer Networks, and it should not be read as "more history hurts". All test exams have at least 10 earlier papers.

7 Bias Check

The data contains no demographic attributes: papers are exam sheets, and instructor names are neither reliable nor used (Deliverable 1). Demographic fairness therefore cannot be assessed. The relevant subgroups are defined by assessment and data coverage: assessment type, subject coverage tier (small, medium or large by the number of training exams of the subject, using tertiles), lab versus theory subjects, target year, and amount of history. For each group Table 15 shows recall, precision and AUC-ROC of the same model as in Section 6 with cluster-bootstrap intervals; Figure 10 shows the same comparison graphically.

Group	Rows	Exams	Recall	Precision	AUC-ROC
Assessment type					
final	192	22	0.831 [0.754, 0.906]	0.607 [0.495, 0.730]	0.875 [0.822, 0.925]
mid	236	29	0.859 [0.779, 0.933]	0.604 [0.509, 0.696]	0.892 [0.861, 0.929]
quiz	140	16	0.590 [0.466, 0.731]	0.679 [0.521, 0.837]	0.734 [0.637, 0.839]
Subject coverage tier					
small	138	19	0.932 [0.872, 1.000]	0.631 [0.508, 0.733]	0.911 [0.857, 0.955]
medium	61	8	0.909 [0.666, 1.000]	0.500 [0.300, 0.684]	0.951 [0.798, 1.000]
large	369	40	0.704 [0.622, 0.785]	0.633 [0.543, 0.718]	0.815 [0.769, 0.862]
Subject kind					
theory	564	66	0.765 [0.699, 0.832]	0.620 [0.553, 0.686]	0.837 [0.794, 0.877]
lab*	4	1	–	–	–
Target year					
2024	210	26	0.791 [0.701, 0.875]	0.673 [0.568, 0.774]	0.846 [0.788, 0.897]
2025	220	26	0.761 [0.639, 0.887]	0.560 [0.452, 0.671]	0.821 [0.753, 0.898]
2026	138	15	0.727 [0.581, 0.892]	0.627 [0.517, 0.750]	0.846 [0.746, 0.936]
Earlier papers of the subject					
10-19	92	13	0.963 [0.862, 1.000]	0.667 [0.514, 0.806]	0.946 [0.852, 0.994]
20+	476	54	0.735 [0.663, 0.808]	0.613 [0.544, 0.686]	0.823 [0.779, 0.864]

Table 15: Metrics of logistic regression on the test split by subgroup, with 95% cluster-bootstrap intervals. *Fewer than 20 rows: not reliable, metrics not shown.

For each grouping we compare the best and the worst group on four metrics (Table 16) and call a gap larger than the uncertainty only if the intervals do not overlap, a conservative test. 4 of the 16 gaps pass it. With this many comparisons, one or two can appear by chance.

Findings. The clearest gap is assessment type: quizzes have a lower recall than mids and finals and a lower AUC-ROC than mids. The gaps between the best group (mids) and the worst (quizzes) exceed the uncertainty for both metrics. Recall also differs by coverage tier (0.932 for small subjects, 0.704 for large ones) and by history, but the precision and AUC-ROC gaps are within the uncertainty, so the recall difference may partly be an effect of the single threshold. Target year shows no separated gap. Lab versus theory cannot be assessed: the test split contains a single lab exam with 4 rows (all other rows are theory subjects), so the result applies to theory subjects.

Data-level bias. The model can only be as fair as its data. Only 28 of 105 subjects have enough history, so results do not describe the other 77 subjects, which are reported as low-confidence and receive no prediction. Scanned papers without extractable text are not spread evenly (Table 17): 51.5% of finals and 44.2% of mids are unreadable, against 21.2% of quizzes. Finals are 23.8% of the clean papers but only 18.2% of the readable ones, whereas quizzes grow from 40.7% to 50.6%.

Grouping	Metric	Best group	Worst group	Gap	Intervals separated
Assessment type	Recall	mid (0.859)	quiz (0.590)	0.269	yes
Assessment type	Precision	quiz (0.679)	mid (0.604)	0.075	no
Assessment type	F1	mid (0.709)	quiz (0.632)	0.078	no
Assessment type	AUC-ROC	mid (0.892)	quiz (0.734)	0.158	yes
Subject coverage tier	Recall	small (0.932)	large (0.704)	0.228	yes
Subject coverage tier	Precision	large (0.633)	medium (0.500)	0.133	no
Subject coverage tier	F1	small (0.752)	medium (0.645)	0.107	no
Subject coverage tier	AUC-ROC	medium (0.951)	large (0.815)	0.136	no
Target year	Recall	2024 (0.791)	2026 (0.727)	0.063	no
Target year	Precision	2024 (0.673)	2025 (0.560)	0.113	no
Target year	F1	2024 (0.727)	2025 (0.646)	0.082	no
Target year	AUC-ROC	2024 (0.846)	2025 (0.821)	0.025	no
Earlier papers of the subject	Recall	10-19 (0.963)	20+ (0.735)	0.228	yes
Earlier papers of the subject	Precision	10-19 (0.667)	20+ (0.613)	0.054	no
Earlier papers of the subject	F1	10-19 (0.788)	20+ (0.668)	0.119	no
Earlier papers of the subject	AUC-ROC	10-19 (0.946)	20+ (0.823)	0.123	no

Table 16: Largest gap between the best and the worst group, per grouping and metric.

The evidence base is therefore thinner for finals. We do not see the expected consequence in the results: finals perform about as well as mids, and the weakest type is quizzes, the type with the most readable papers. This does not prove that the scan imbalance is harmless; the cause of the quiz weakness is untested. Quizzes make up 21.0% of the training rows, 14.0% of the validation rows and 24.6% of the test rows, so they are not underrepresented in the evaluation data.

Type	Papers	Readable	Unreadable	Unreadable share	Share of all papers	Share of readable papers
quiz	1171	923	248	21.2%	40.7%	50.6%
mid	1010	564	446	44.2%	35.1%	30.9%
final	684	332	352	51.5%	23.8%	18.2%
unspecified	9	4	5	55.6%	0.3%	0.2%
All	2874	1823	1051	36.6%	100%	100%

Table 17: Readable text by assessment type among the 2,874 clean papers.

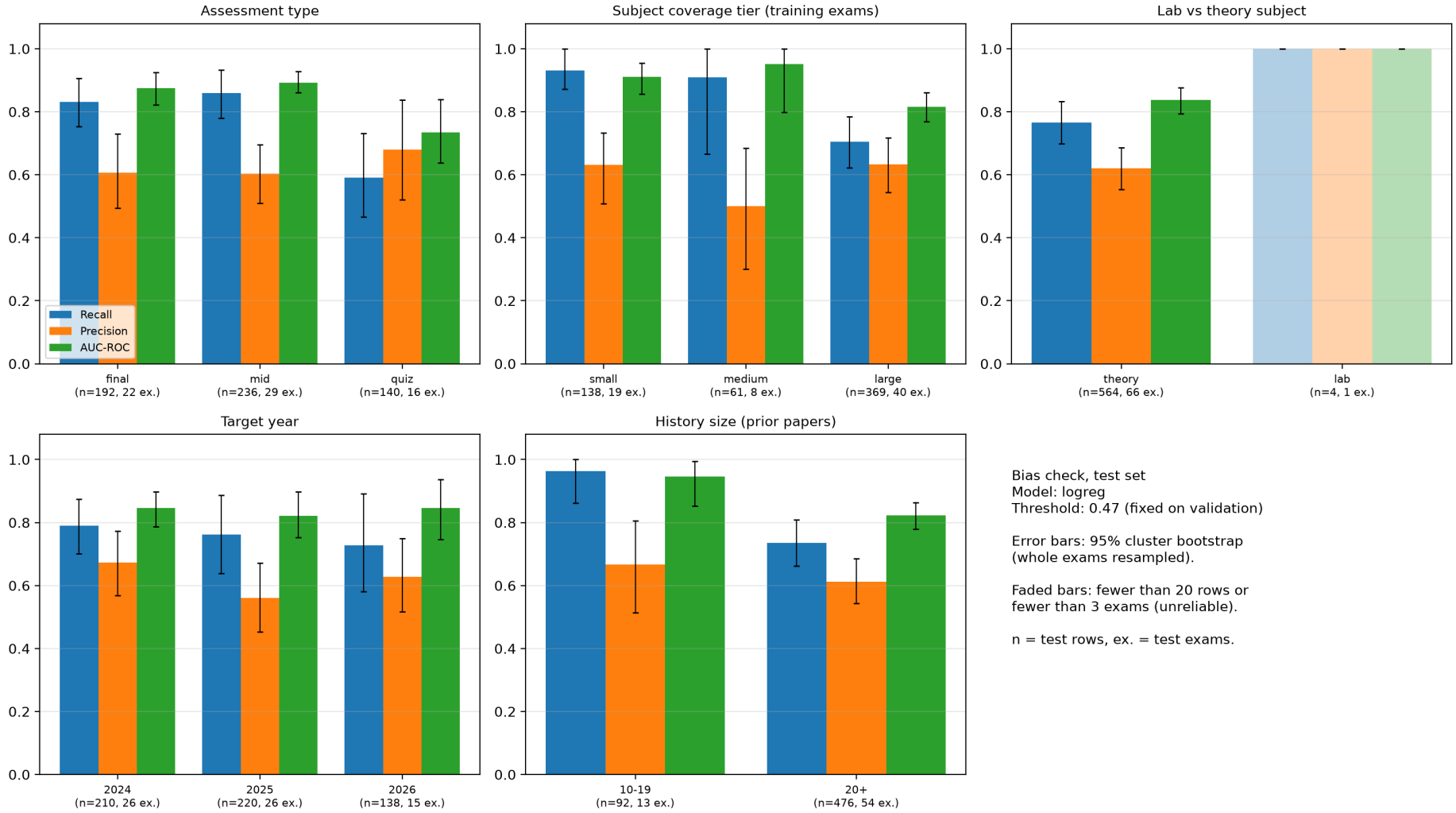


Figure 10: Metrics of logistic regression by subgroup on the test split (error bars: 95% cluster-bootstrap intervals; faded bars: fewer than 20 rows).

8 Limitations

- **Topics are approximate.** They come from unsupervised clustering with low silhouette scores, and some clusters capture generic wording. Labels inherit this noise. Question-level topic labelling with NLP remains future work.
- **The label depends on group size.** An exam group made of several papers (for example Mid 1 and Mid 2 of one year) has more chances to contain a topic than a single quiz. We do not add group size as a feature because it would use target-year information.
- **Small evaluation sets.** Validation has 36 exams and test 67, so metric differences of a few points are noise, and the best of 40 tuned configurations is optimistic on validation.
- **Distribution drift.** The positive rate falls from 47.3% to 39.2% and 34.7%, and the predicted probabilities are too high.
- **Limited coverage.** 28 of 105 subjects are covered, and the test split contains almost only theory subjects. More than a third of the clean papers (36.6%) have no readable text and need OCR, so papers and subjects without readable text are missing.
- **No cold-start measurement.** The test split has no never-seen topics, and rare topics have only 8 positives.
- **Weak baseline gap.** A one-line history rule is as good as the learned models in ranking, so the benefit of learning on these features is not demonstrated.
- **Metadata and segmentation were checked by sampling, not measured** against a manually labelled set.
- **Source quality.** The archive is student-contributed and may contain mislabelled or incomplete papers; it is not the current syllabus.

9 Reproducibility, Repository and Model Weights

Repository. https://github.com/Minahilbasalat/SpillThePaper. Phase 3 and 4 code is in `src/data/` (task, questions, topics, dataset), `src/models/` (preprocessing, baselines, MLP, tuning, prediction export) and `src/evaluation/` (metrics, baselines, evaluation, error analysis, bias check, contribution table). Tests are in `tests/`, committed results in `results/`, figures in `report/figures/`, and documentation in `docs/`. All work was committed with the Git command line, and every member's commits are listed in Section 10.

Environment. `requirements.txt` lists the dependencies (pandas, matplotlib, seaborn, PyYAML, PyMuPDF, python-docx, requests, scikit-learn, torch, joblib, pytest). Python 3.14, scikit-learn 1.9.1 and PyTorch 2.15.0 (CPU) were used. Create a virtual environment, activate it and run `pip install -r requirements.txt`.

Reproduce (from the repository root, with the environment active; the committed tables in `data/processed/model/` are the inputs, so the Deliverable 1 download is not repeated):

```
python -m src.models.baselines
python -m src.models.tune
python -m src.models.train_mlp
python -m src.models.export_predictions
python -m src.models.plots
python -m src.evaluation.baselines
python -m src.evaluation.evaluate
```

```
python -m src.evaluation.error_analysis
python -m src.evaluation.bias_check
python -m src.evaluation.contribution_table
python -m pytest tests -q
```

The data-side scripts (`src/data/`) need the raw archive of Deliverable 1 and are documented in the README; their outputs are committed.

Model weights. Weight files are not stored in Git. Following the submission rule, they are hosted externally: Google Drive folder:

`https://drive.google.com/drive/folders/1IQb9xincr7PjG4kQvqEmj8jBBezRqQRZ?usp=drive_link`

The folder contains `mlp_best.pt` (the final MLP), `preprocess.joblib` (the preprocessing fitted on the training split), `logreg.joblib` and `random_forest.joblib`. The files are small (a few kilobytes for the MLP, about 1.4 MB for the random forest). They are the outputs of the training scripts above, run on the committed data. All reported metrics are computed from the committed prediction files in `results/predictions/`. Re-running the training scripts on a second machine reproduced all six committed prediction files (validation and test, three models) exactly, with a maximum absolute score difference of 0.

10 Contribution Table

Table 18 maps Git commit authors to project modules for the Phase 3 and 4 work (commits on `main`, merge commits excluded). It was produced from the Git history by `src/evaluation/contribution_table.py` after the code and the evaluation were finished, so the final documentation commits (README and report) are not counted. Author names and e-mail aliases were mapped to the member's real name by e-mail address. The history contains 60 such commits: 31 by Minahil Basalat, 18 by Khadija Rao and 11 by Iman Abid. A commit that touches several modules counts once for each. "Python lines" counts added lines in `.py` files of the module.

The `results` and `report` rows of the first member record outputs (tables and figures) that the evaluation scripts generated and that were committed during integration; the code that produced them is in `src/evaluation`. The skeleton commit also created empty package files in `src/models` and `src/evaluation`. Table 19 lists selected commits for each member.

Member	Module	Commits	Python lines	Contents
Minahil Basalat	`src/data`	12	674	schema constants, text loader, question segmenter, topic discovery, dataset builder, dummy-data generator, dataset figure
	`tests`	4	92	segmenter tests; leakage test of the dataset builder
	`docs`	3	–	dataset schema; data-side settings
	`results`	6	–	Phase 4 outputs (predictions of the rule baselines, metrics, error-analysis and bias tables) produced by the evaluation scripts and committed during integration
	`report`	4	–	dataset figure; evaluation figures produced by the evaluation scripts, committed during integration
	`other`	5	–	package skeleton, `requirements.txt`, `.gitignore`, README
Khadija Rao	`src/models`	9	844	preprocessing, baselines, MLP, training loop, random search, prediction export, figure generation
	`tests`	1	137	preprocessing tests (train-only fitting, no missing values)
	`docs`	2	–	model documentation and weights link
	`results`	5	–	baseline and tuning results, MLP training history, predictions of the three learned models
	`report`	2	–	architecture, training-curve and tuning figures
Iman Abid	`src/evaluation`	9	1435	metrics, cluster bootstrap, predictions contract, rule baselines, evaluation, error analysis, bias check, contribution table
	`tests`	1	92	metric tests against hand-computed examples
	`docs`	1	–	evaluation documentation

Table 18: Contribution of each member by module (Phase 3 and 4, from the Git history of `main`).

Author	Commit	Module	Message
Minahil Basalat	`ed0e3ac`	`src/data`	feat(data): add loader that joins papers_clean with extracted text
Minahil Basalat	`268874c`	`src/data`	feat(data): add rule-based question segmenter with marker, numbered and paragraph fallbacks
Minahil Basalat	`0135a7a`	`data files, src/data`	feat(data): per-subject topic discovery (TF-IDF + KMeans) fitted on train years only
Minahil Basalat	`c83ce19`	`src/data`	feat(data): add leakage-safe topic-appearance dataset builder with time-based split
Minahil Basalat	`0334616`	`tests`	test(data): verify dataset features use only earlier years (leakage test)
Khadija Rao	`ce3d232`	`src/models`	feat(models): add train-only preprocessing (imputation, one-hot, scaling)
Khadija Rao	`3be7d1a`	`src/models`	feat(models): add logistic regression and random forest baselines tuned on val AUC
Khadija Rao	`b947601`	`src/models`	feat(models): add PyTorch MLP with BCEWithLogits loss and Adam/SGD options
Khadija Rao	`87de66c`	`src/models`	feat(models): add random search over MLP hyperparameters (40 configs x 3 seeds)
Khadija Rao	`d6516d1`	`src/models`	feat(models): add MLP training loop with early stopping and history logging
Iman Abid	`b15d35e`	`src/evaluation`	feat(evaluation): add metrics, F1 threshold selection and cluster bootstrap
Iman Abid	`b42fdb9`	`src/evaluation`	feat(evaluation): add rule-based baselines in the shared predictions format
Iman Abid	`a8e4b7f`	`src/evaluation`	feat(evaluation): add evaluate script with val-chosen threshold and cluster-bootstrap CIs
Iman Abid	`0954390`	`src/evaluation`	feat(evaluation): add error analysis by group with top false negatives and positives
Iman Abid	`5dab161`	`src/evaluation`	feat(evaluation): add bias check across assessment, coverage, lab/theory, year and history groups

Table 19: Selected commits (full history: `results/contribution_commits.csv`).

11 Conclusion

We turned an unlabelled archive of exam papers into a supervised, leakage-safe task, trained three models with documented tuning, and evaluated them chronologically with confidence intervals, error analysis and a bias check. The learned models rank topics with an AUC-ROC of about 0.84 to 0.85 and are clearly more precise than the "appeared five times" rule of Deliverable 1, but they are not demonstrably better than "appeared last year". They are weakest on quizzes and on Computer Networks, overestimate probabilities, and cannot be evaluated on topics never seen before. The main next steps are OCR for the 36.6% of clean papers without readable text, question-level topic labelling with NLP and comparison with the current syllabus, a model that tolerates the missing cold-start case, and, in later phases, the tentative paper generator and the personalised practice.